2520090173 – MEKONDA ABHI SAI SAGAR WEEK 2 – SKILL

Experiment Summary

Aim: To analyse packet captures taken from three points in a network path (sender host1, router, receiver host2) in order to understand packet flow and identify performance differences. A virtual network of two hosts connected through a router is built with Linux network namespaces and veth pairs, and the same traffic is captured simultaneously at all three points with tshark.

Network topology:

Link	End A (IP / Interface)	End B (IP / Interface)
Link 1	host1 - 10.0.1.1/30 / veth-h1	router - 10.0.1.2/30 / veth-r1
Link 2	router - 10.0.2.1/30 / veth-r2	host2 - 10.0.2.2/30 / veth-h2

host1 sends to host2 (10.0.2.2). host1 reaches the 10.0.2.0/30 network via the router at 10.0.1.2, and host2 reaches the 10.0.1.0/30 network via the router at 10.0.2.1.

Stages (key commands)

Stage 1 - Install tools:

```
sudo apt update
sudo apt install -y tshark iperf3 mtr-tiny iproute2 wireshark netcat-openbsd
```

Stage 2 - Build the network (namespaces and veth pairs):

```
sudo ip netns add host1 / router / host2
sudo ip link add veth-h1 type veth peer name veth-r1
sudo ip link add veth-r2 type veth peer name veth-h2
Move each interface into its namespace (ip link set <iface> netns <node>); verify: sudo ip netns list
```

Stage 3 - Configure IP addresses:

Assign the IPs from the topology table and bring every interface (and lo) up.

Stage 4 - Enable routing:

```
sudo ip netns exec router sysctl -w net.ipv4.ip_forward=1
sudo ip netns exec host1 ip route add 10.0.2.0/30 via 10.0.1.2
sudo ip netns exec host2 ip route add 10.0.1.0/30 via 10.0.2.1
```

Stage 5 - Test connectivity:

```
sudo ip netns exec host1 ping -c 3 10.0.2.2
```

Stage 6 - Add delay and loss on the router:

```
sudo ip netns exec router tc qdisc add dev veth-r2 root netem delay 20ms loss 1%
sudo ip netns exec router tc qdisc show dev veth-r2   (verify the rule)
```

Stage 7-8 - Start three captures simultaneously (separate terminals, leave running):

```
sudo ip netns exec host1 tshark -i veth-h1 -w /tmp/cap1.pcap
sudo ip netns exec router tshark -i veth-r1 -w /tmp/cap2.pcap
sudo ip netns exec host2 tshark -i veth-h2 -w /tmp/cap3.pcap
```

Stage 9 - Send a real message (netcat):

```
Terminal 4 (receiver):  sudo ip netns exec host2 nc -l 1234
```

```
Terminal 5 (sender):    sudo ip netns exec host1 nc 10.0.2.2 1234, then
type the messages
```

Stage 10-11 - Stop captures, fix permissions, check files:

```
Ctrl+C in all terminals
sudo chmod 644 /tmp/cap1.pcap /tmp/cap2.pcap /tmp/cap3.pcap
ls -lh /tmp/cap1.pcap /tmp/cap2.pcap /tmp/cap3.pcap   (all sizes > 0)
```

Stage 12-14 - Analysis:

Check the TTL in each capture (tshark -r <file> -T fields -e ip.ttl), read the message back from cap2 (tcp.port == 1234), compare the first SYN time in cap1 and cap3, count retransmissions in each file, run mtr --report, and merge the captures in Wireshark with a TTL column.

Clean up:

```
sudo ip netns del host1 / router / host2;  sudo ip netns list  (must
return blank)
```

Results (Expected Outcomes)

The following are the expected outcomes of the procedure above, as stated in the lab guide; they are the values the three captures must show when the experiment is performed correctly.

Connectivity: the ping test from host1 to host2 reports 3 packets transmitted, 3 received, 0% packet loss, proving end-to-end routing through the router namespace.

TTL at the three capture points: cap1 = 64, cap2 = 64, cap3 = 63. Linux hosts originate packets with a TTL of 64, so the packet is seen with TTL 64 at host1 and again at the router's incoming interface, because nothing has forwarded it yet. When the router forwards the packet, it decrements the TTL by 1 (a router always reduces TTL by 1 for every hop it forwards), so the packet arrives at host2, and is captured in cap3, with TTL 63. The difference of exactly 1 is direct proof that the packet crossed one routed hop.

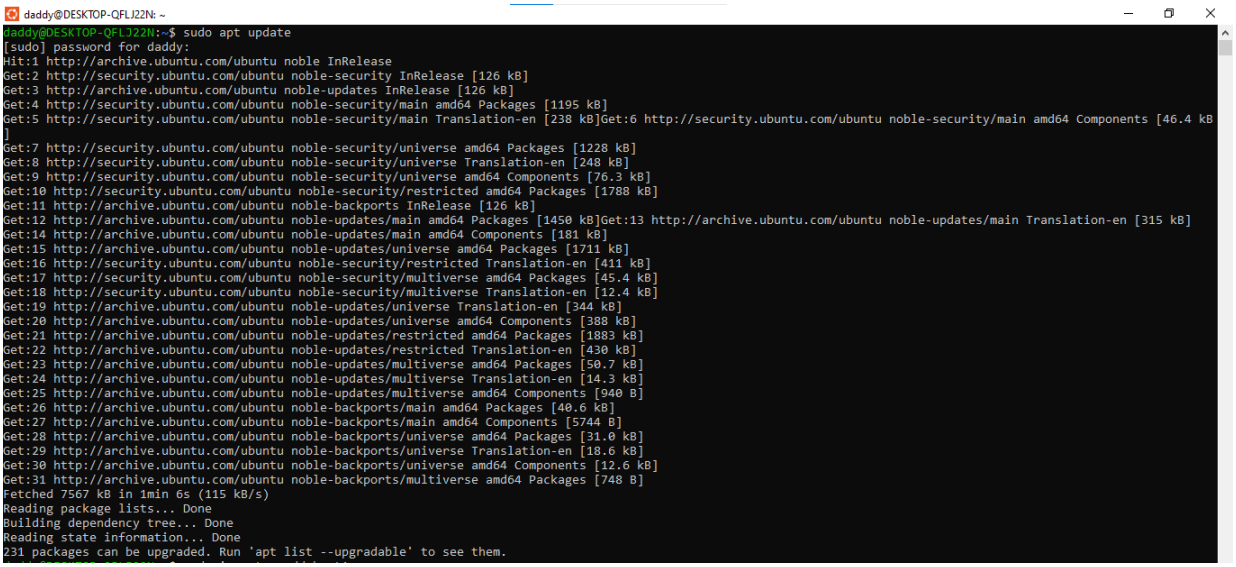
Delay: the first SYN packet appears in cap3 approximately 20 ms later than in cap1, matching the netem rule delay 20ms loss 1% applied on the router's veth-r2 interface.

Loss and retransmissions: the retransmission count is highest at cap3, because the 1% loss was applied on the router-to-host2 link (veth-r2) - after the router capture point and before the host2 capture point - so packets dropped on that last link are missing at host2 and TCP retransmits them.

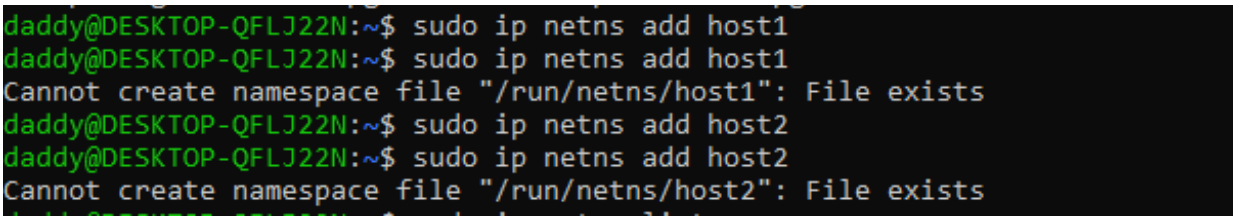
Message delivery: the message typed at host1 (starting with "Hello from host1!") is received and displayed on host2 by the netcat receiver listening on port 1234 (nc -l 1234), and the same message text can be read back from the TCP payload of the router capture (cap2).

Screenshots

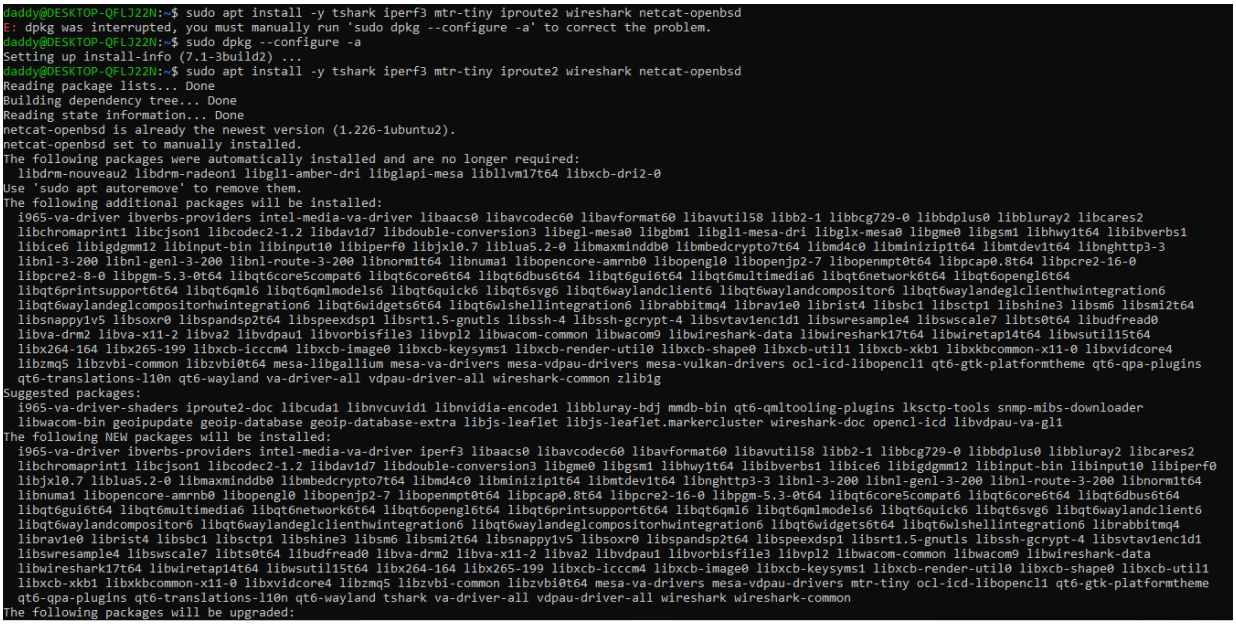
Terminal 1 – Full run from the start (all stages in order):



Terminal 1 - Step 1: sudo apt update



Terminal 1 - Creating namespaces: host1 and host2 already existed from an earlier attempt



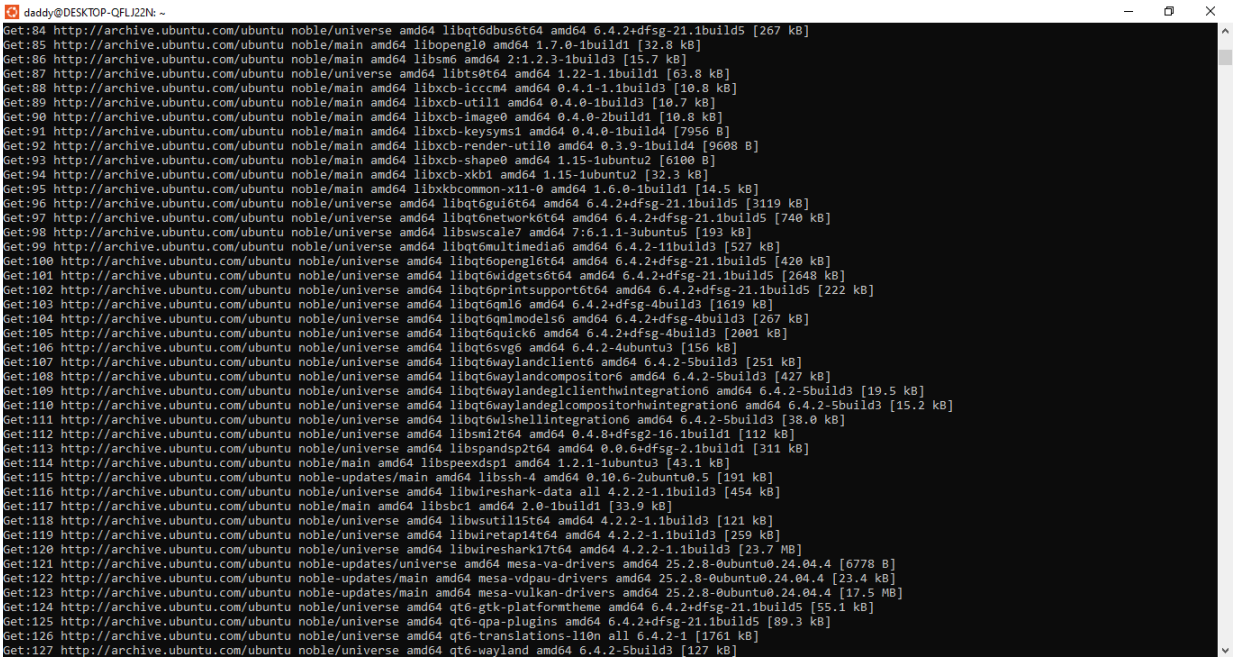
Terminal 1 - Installing tools: dpkg repair (sudo dpkg --configure -a), then sudo apt install -y tshark iperf3 mtr-tiny iproute2 wireshark netcat-openbsd

```
The following packages will be upgraded:
  iproute2 libegl-mesa0 libgbm1 libgl1-mesa-dri libglx-mesa0 libpcre2-8-0 libssh-4 mesa-libgallium mesa-vulkan-drivers zlib1g
10 upgraded, 123 newly installed, 0 to remove and 221 not upgraded.
Need to get 111 MB of archives.
After this operation, 323 MB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu noble/main amd64 libsctp1 amd64 1.0.19+dfsg-2build1 [9146 B]
Get:2 http://archive.ubuntu.com/ubuntu noble/universe amd64 libiperf0 amd64 3.16-1build2 [87.1 kB]
Get:3 http://archive.ubuntu.com/ubuntu noble/universe amd64 iperf3 amd64 3.16-1build2 [19.0 kB]
Get:4 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libpcre2-8-0 amd64 10.42-4ubuntu2.2 [229 kB]
Get:5 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 zlib1g amd64 1:1.3.dfsg-3.1ubuntu2.3 [63.2 kB]
Get:6 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 iproute2 amd64 6.1.0-1ubuntu6.4 [1120 kB]
Get:7 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libnl-3-200 amd64 3.7.0-0.3build1.1 [55.7 kB]
Get:8 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libnl-route-3-200 amd64 3.7.0-0.3build1.1 [189 kB]
Get:9 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libibverbs1 amd64 50.0-2ubuntu0.2 [68.0 kB]
Get:10 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 ibverbs-providers amd64 50.0-2ubuntu0.2 [381 kB]
Get:11 http://archive.ubuntu.com/ubuntu noble/main amd64 libmaxminddb0 amd64 1.9.1-1build1 [24.4 kB]
Get:12 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libnuma1 amd64 2.0.18-1ubuntu0.24.04.1 [23.4 kB]
Get:13 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libpcap0.8t64 amd64 1.10.4-4.1ubuntu3.1 [151 kB]
Get:14 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 mtr-tiny amd64 0.95-1.1ubuntu0.1 [54.3 kB]
Get:15 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 libva2 amd64 2.20.0-2ubuntu0.2 [66.4 kB]
Get:16 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 libigdgmm12 amd64 22.3.17+ds1-1ubuntu1 [145 kB]
Get:17 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 intel-media-va-driver amd64 24.1.0+dfsg1-1ubuntu0.2 [3163 kB]
Get:18 http://archive.ubuntu.com/ubuntu noble/universe amd64 libaacs0 amd64 0.11.1-2build1 [62.9 kB]
Get:19 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 libva-drm2 amd64 2.20.0-2ubuntu0.2 [7124 B]
Get:20 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 libva-x11-2 amd64 2.20.0-2ubuntu0.2 [12.0 kB]
Get:21 http://archive.ubuntu.com/ubuntu noble/main amd64 libvdpau1 amd64 1.5-2build1 [27.8 kB]
Get:22 http://archive.ubuntu.com/ubuntu noble/universe amd64 libvpl2 amd64 2023.3.0-1build1 [99.8 kB]
Get:23 http://archive.ubuntu.com/ubuntu noble/universe amd64 ocl-icd-libopencl1 amd64 2.3.2-1build1 [38.5 kB]
Get:24 http://archive.ubuntu.com/ubuntu noble/universe amd64 libavutil58 amd64 7:6.1.1-3ubuntu5 [401 kB]
Get:25 http://archive.ubuntu.com/ubuntu noble/universe amd64 libcodec2-1.2 amd64 1.2.0-2build1 [8998 kB]
Get:26 http://archive.ubuntu.com/ubuntu noble/universe amd64 libdav1d7 amd64 1.4.1-1build1 [604 kB]
Get:27 http://archive.ubuntu.com/ubuntu noble/universe amd64 libgsm1 amd64 1.0.22-1build1 [27.8 kB]
Get:28 http://archive.ubuntu.com/ubuntu noble/universe amd64 libhwy1t64 amd64 1.0.7-8.1build1 [584 kB]
Get:29 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 libjxl0.7 amd64 0.7.0-10.2ubuntu6.1 [1001 kB]
Get:30 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libopenjp2-7 amd64 2.5.0-2ubuntu0.5 [174 kB]
Get:31 http://archive.ubuntu.com/ubuntu noble/universe amd64 librav1e0 amd64 0.7.1-2 [1022 kB]
Get:32 http://archive.ubuntu.com/ubuntu noble/universe amd64 libshine3 amd64 3.1.1-2build1 [23.2 kB]
Get:33 http://archive.ubuntu.com/ubuntu noble/main amd64 libsnappy1v5 amd64 1.1.10-1build1 [28.6 kB]
Get:34 http://archive.ubuntu.com/ubuntu noble/universe amd64 libsvtav1enc1d1 amd64 1.7.0+dfsg-2build1 [2425 kB]
Get:35 http://archive.ubuntu.com/ubuntu noble/universe amd64 libsoxr0 amd64 0.1.3-4build3 [80.0 kB]
Get:36 http://archive.ubuntu.com/ubuntu noble/universe amd64 libswresample4 amd64 7:6.1.1-3ubuntu5 [63.8 kB]
Get:37 http://archive.ubuntu.com/ubuntu noble/universe amd64 libx264-164 amd64 2:0.164.3108+git31e19f9-1 [604 kB]
Get:38 http://archive.ubuntu.com/ubuntu noble/universe amd64 libx265-199 amd64 3.5-2build1 [1226 kB]
Get:39 http://archive.ubuntu.com/ubuntu noble/universe amd64 libxvidcore4 amd64 2:1.3.7-1build1 [219 kB]
```

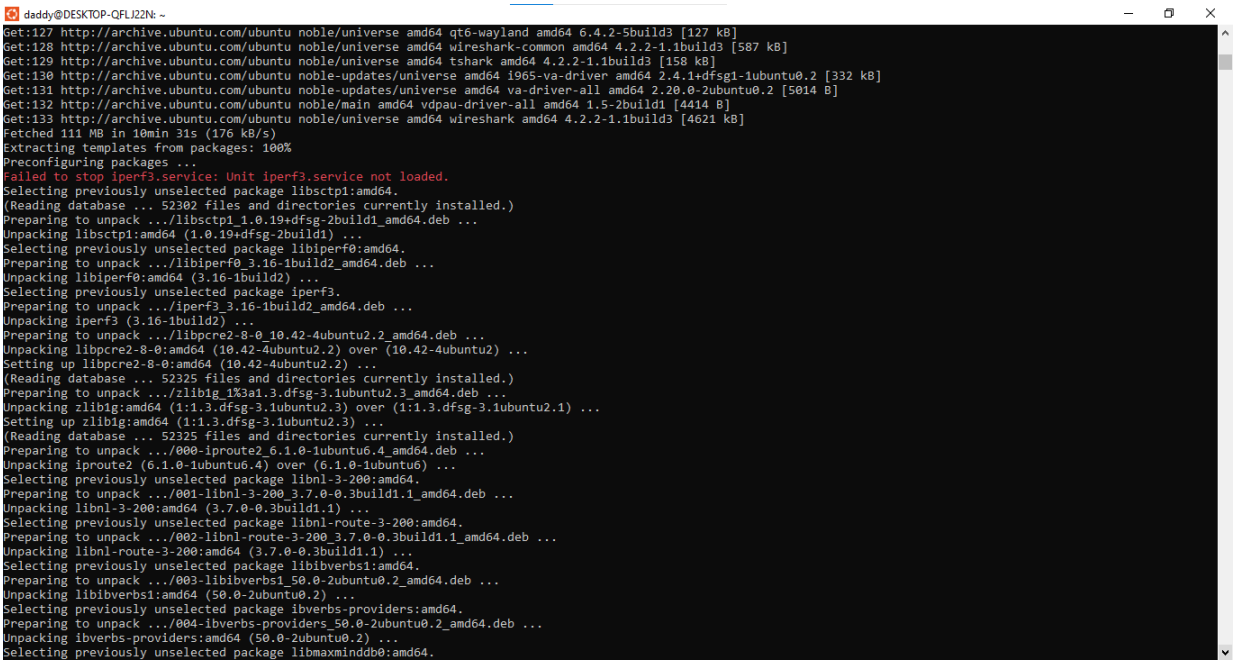
Terminal 1 - Downloading packages (1 of 3)

```
daddy@DESKTOP-QFLJ22N: ~
Get:40 http://archive.ubuntu.com/ubuntu noble/universe amd64 libzvbi-common all 0.2.42-2 [42.4 kB]
Get:41 http://archive.ubuntu.com/ubuntu noble/universe amd64 libzvbi0t64 amd64 0.2.42-2 [261 kB]
Get:42 http://archive.ubuntu.com/ubuntu noble/universe amd64 libavcodec60 amd64 7:6.1.1-3ubuntu5 [5851 kB]
Get:43 http://archive.ubuntu.com/ubuntu noble/universe amd64 libudfread0 amd64 1.1.2-1build1 [19.0 kB]
Get:44 http://archive.ubuntu.com/ubuntu noble/universe amd64 libbluray2 amd64 1:1.3.4-1build1 [159 kB]
Get:45 http://archive.ubuntu.com/ubuntu noble/universe amd64 libchromaprint1 amd64 1.5.1-5 [30.5 kB]
Get:46 http://archive.ubuntu.com/ubuntu noble/universe amd64 libgme0 amd64 0.6.3-7build1 [134 kB]
Get:47 http://archive.ubuntu.com/ubuntu noble/main amd64 libvorbisfile3 amd64 1.3.7-1build3 [17.1 kB]
Get:48 http://archive.ubuntu.com/ubuntu noble/universe amd64 libopenmpt0t64 amd64 0.7.3-1.1build3 [647 kB]
Get:49 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 librabbitmq4 amd64 0.11.0-1ubuntu0.2 [38.3 kB]
Get:50 http://archive.ubuntu.com/ubuntu noble/universe amd64 libcjson1 amd64 1.7.17-1 [24.8 kB]
Get:51 http://archive.ubuntu.com/ubuntu noble/universe amd64 libmbedcrypto7t64 amd64 2.28.8-1 [209 kB]
Get:52 http://archive.ubuntu.com/ubuntu noble/universe amd64 librist4 amd64 0.2.10+dfsg-2 [74.9 kB]
Get:53 http://archive.ubuntu.com/ubuntu noble/universe amd64 libsrt1.5-gnutls amd64 1.5.3-1build2 [316 kB]
Get:54 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libssh-gcrypt-4 amd64 0.10.6-2ubuntu0.5 [226 kB]
Get:55 http://archive.ubuntu.com/ubuntu noble/universe amd64 libnorm1t64 amd64 1.5.9+dfsg-3.1build1 [154 kB]
Get:56 http://archive.ubuntu.com/ubuntu noble/universe amd64 libpgm-5.3-0t64 amd64 5.3.128~dfsg-2.1build1 [167 kB]
Get:57 http://archive.ubuntu.com/ubuntu noble/universe amd64 libzmq5 amd64 4.3.5-1build2 [260 kB]
Get:58 http://archive.ubuntu.com/ubuntu noble/universe amd64 libavformat60 amd64 7:6.1.1-3ubuntu5 [1153 kB]
Get:59 http://archive.ubuntu.com/ubuntu noble/universe amd64 libb2-1 amd64 0.98.1-1.1build1 [45.0 kB]
Get:60 http://archive.ubuntu.com/ubuntu noble/universe amd64 libbcg729-0 amd64 1.1.1-2build1 [35.3 kB]
Get:61 http://archive.ubuntu.com/ubuntu noble/universe amd64 libbdplus0 amd64 0.2.0-3build1 [52.2 kB]
Get:62 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libcares2 amd64 1.27.0-1.0ubuntu1.1 [74.2 kB]
Get:63 http://archive.ubuntu.com/ubuntu noble/universe amd64 libdouble-conversion3 amd64 3.3.0-1build1 [40.3 kB]
Get:64 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libgl1-mesa-dri amd64 25.2.8-0ubuntu0.24.04.4 [38.0 kB]
Get:65 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libglx-mesa0 amd64 25.2.8-0ubuntu0.24.04.4 [110 kB]
Get:66 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libegl-mesa0 amd64 25.2.8-0ubuntu0.24.04.4 [117 kB]
Get:67 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libgbm1 amd64 25.2.8-0ubuntu0.24.04.4 [34.2 kB]
Get:68 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 mesa-libgallium amd64 25.2.8-0ubuntu0.24.04.4 [10.8 MB]
Get:69 http://archive.ubuntu.com/ubuntu noble/main amd64 libice6 amd64 2:1.0.10-1build3 [41.4 kB]
Get:70 http://archive.ubuntu.com/ubuntu noble/main amd64 libwacom-common all 2.10.0-2 [63.4 kB]
Get:71 http://archive.ubuntu.com/ubuntu noble/main amd64 libwacom9 amd64 2.10.0-2 [23.9 kB]
Get:72 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libinput-bin amd64 1.25.0-1ubuntu3.7 [23.6 kB]
Get:73 http://archive.ubuntu.com/ubuntu noble/main amd64 libmtdev1t64 amd64 1.1.6-1.1build1 [14.4 kB]
Get:74 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libinput10 amd64 1.25.0-1ubuntu3.7 [133 kB]
Get:75 http://archive.ubuntu.com/ubuntu noble/universe amd64 liblua5.2-0 amd64 5.2.4-3build2 [123 kB]
Get:76 http://archive.ubuntu.com/ubuntu noble/universe amd64 libmd4c0 amd64 0.4.8-1build1 [42.3 kB]
Get:77 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 libminizip1t64 amd64 1:1.3.dfsg-3.1ubuntu2.3 [22.2 kB]
Get:78 http://archive.ubuntu.com/ubuntu noble/universe amd64 libnghttp3-3 amd64 0.8.0-2 [59.0 kB]
Get:79 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libnl-genl-3-200 amd64 3.7.0-0.3build1.1 [12.2 kB]
Get:80 http://archive.ubuntu.com/ubuntu noble/universe amd64 libopencore-amrnb0 amd64 0.1.6-1build1 [98.2 kB]
Get:81 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libpcre2-16-0 amd64 10.42-4ubuntu2.2 [212 kB]
Get:82 http://archive.ubuntu.com/ubuntu noble/universe amd64 libqt6core6t64 amd64 6.4.2+dfsg-21.1build5 [1789 kB]
Get:83 http://archive.ubuntu.com/ubuntu noble/universe amd64 libqt6core5compat6 amd64 6.4.2-4build3 [147 kB]
```

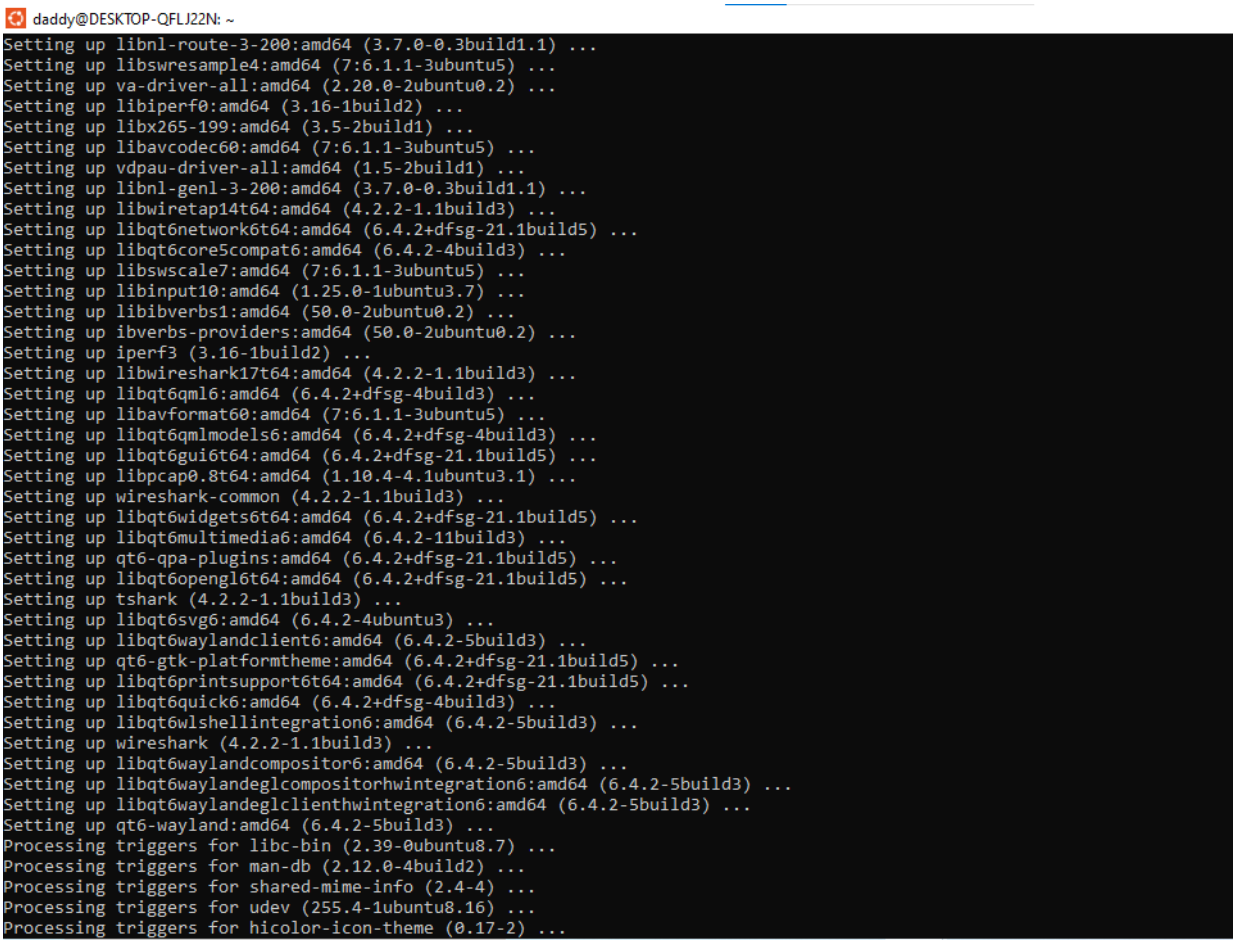
Terminal 1 - Downloading packages (2 of 3)



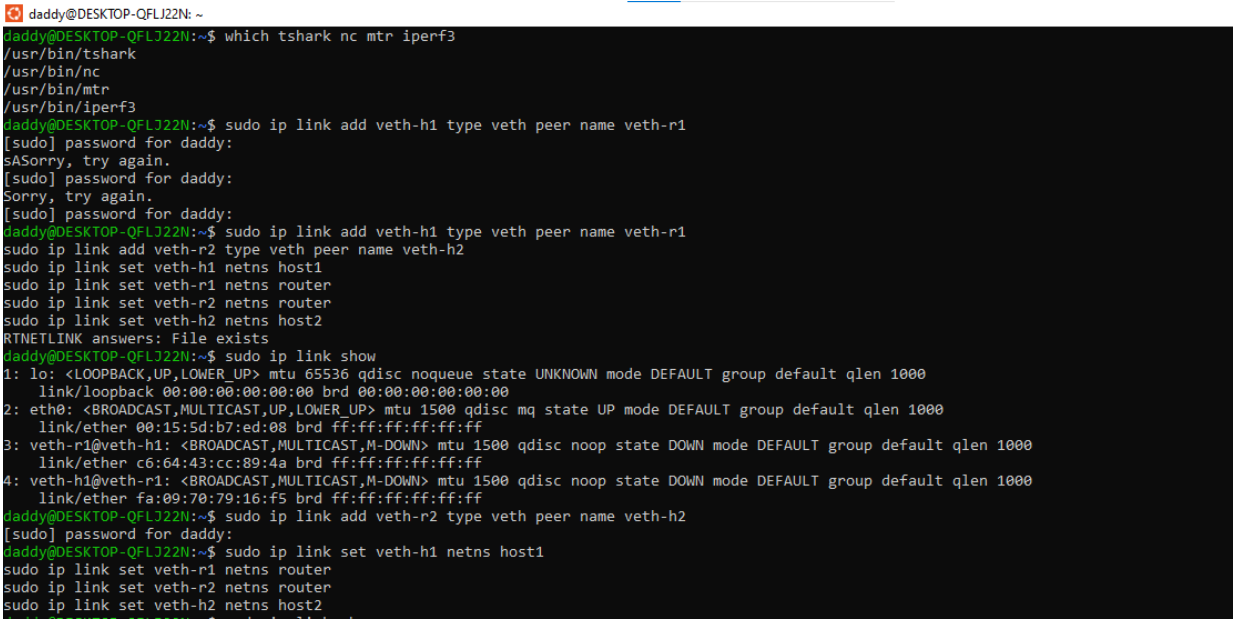
Terminal 1 - Downloading packages (3 of 3)



Terminal 1 - Download complete (111 MB fetched), unpacking packages



Terminal 1 - Tools installed: tshark, wireshark and iperf3 set up



Terminal 1 - Tool check (which tshark nc mtr iperf3) and veth pairs created

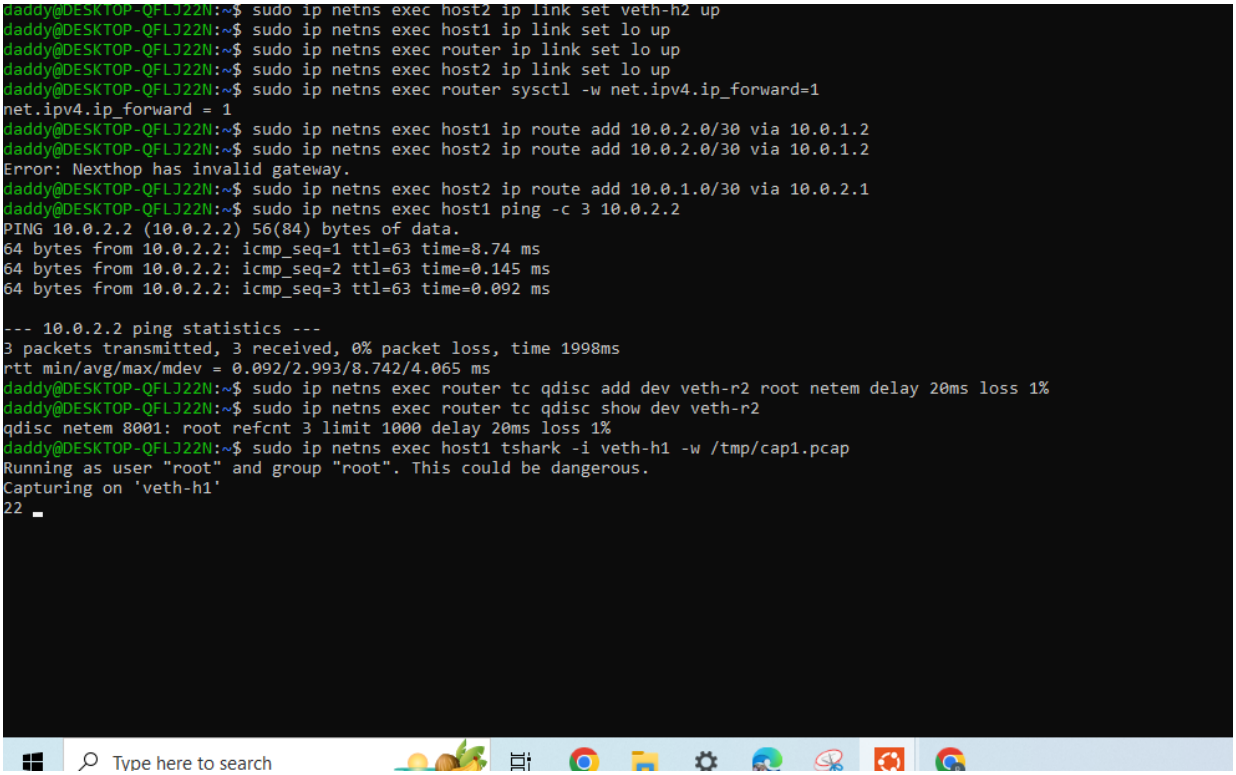
```
daddy@DESKTOP-QFLJ22N:~$ sudo ip link show
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN mode DEFAULT group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc mq state UP mode DEFAULT group default qlen 1000
    link/ether 00:15:5d:b7:ed:08 brd ff:ff:ff:ff:ff:ff
3: veth-r1@veth-h1: <BROADCAST,MULTICAST,M-DOWN> mtu 1500 qdisc noop state DOWN mode DEFAULT group default qlen 1000
    link/ether c6:64:43:cc:89:4a brd ff:ff:ff:ff:ff:ff
4: veth-h1@veth-r1: <BROADCAST,MULTICAST,M-DOWN> mtu 1500 qdisc noop state DOWN mode DEFAULT group default qlen 1000
    link/ether fa:09:70:79:16:f5 brd ff:ff:ff:ff:ff:ff
daddy@DESKTOP-QFLJ22N:~$ sudo ip link add veth-r2 type veth peer name veth-h2
[sudo] password for daddy:
daddy@DESKTOP-QFLJ22N:~$ sudo ip link set veth-h1 netns host1
sudo ip link set veth-r1 netns router
sudo ip link set veth-r2 netns router
sudo ip link set veth-h2 netns host2
daddy@DESKTOP-QFLJ22N:~$ sudo ip link show
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN mode DEFAULT group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc mq state UP mode DEFAULT group default qlen 1000
    link/ether 00:15:5d:b7:ed:08 brd ff:ff:ff:ff:ff:ff
3: veth-r1@if4: <BROADCAST,MULTICAST> mtu 1500 qdisc noop state DOWN mode DEFAULT group default qlen 1000
    link/ether c6:64:43:cc:89:4a brd ff:ff:ff:ff:ff:ff link-netns host1
5: veth-h2@veth-r2: <BROADCAST,MULTICAST,M-DOWN> mtu 1500 qdisc noop state DOWN mode DEFAULT group default qlen 1000
    link/ether ea:58:f1:a6:6a:ad brd ff:ff:ff:ff:ff:ff
6: veth-r2@veth-h2: <BROADCAST,MULTICAST,M-DOWN> mtu 1500 qdisc noop state DOWN mode DEFAULT group default qlen 1000
    link/ether 8e:4b:92:53:8c:60 brd ff:ff:ff:ff:ff:ff
daddy@DESKTOP-QFLJ22N:~$ sudo ip link set veth-r1 netns router
daddy@DESKTOP-QFLJ22N:~$ sudo ip link set veth-r2 netns router
daddy@DESKTOP-QFLJ22N:~$ sudo ip link set veth-h2 netns host2
daddy@DESKTOP-QFLJ22N:~$ sudo ip link show
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN mode DEFAULT group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc mq state UP mode DEFAULT group default qlen 1000
    link/ether 00:15:5d:b7:ed:08 brd ff:ff:ff:ff:ff:ff
daddy@DESKTOP-QFLJ22N:~$ sudo ip netns exec host1 ip addr add 10.0.1.1/30 dev veth-h1
sudo ip netns exec router ip addr add 10.0.1.2/30 dev veth-r1
sudo ip netns exec router ip addr add 10.0.2.1/30 dev veth-r2
sudo ip netns exec host2 ip addr add 10.0.2.2/30 dev veth-h2
```

Terminal 1 - veth interfaces moved into host1, router and host2 namespaces

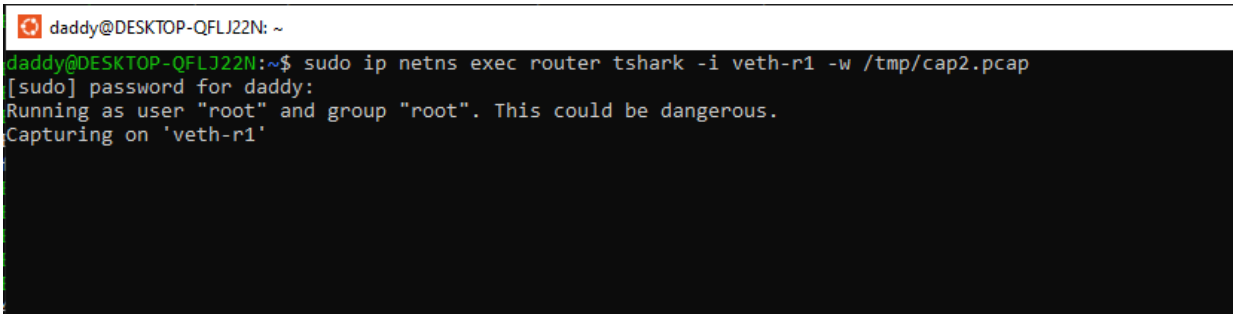
```
daddy@DESKTOP-QFLJ22N: ~
    link/ether 8e:4b:92:53:8c:60 brd ff:ff:ff:ff:ff:ff
daddy@DESKTOP-QFLJ22N:~$ sudo ip link set veth-r1 netns router
daddy@DESKTOP-QFLJ22N:~$ sudo ip link set veth-r2 netns router
daddy@DESKTOP-QFLJ22N:~$ sudo ip link set veth-h2 netns host2
daddy@DESKTOP-QFLJ22N:~$ sudo ip link show
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN mode DEFAULT group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc mq state UP mode DEFAULT group default qlen 1000
    link/ether 00:15:5d:b7:ed:08 brd ff:ff:ff:ff:ff:ff
daddy@DESKTOP-QFLJ22N:~$ sudo ip netns exec host1 ip addr add 10.0.1.1/30 dev veth-h1
sudo ip netns exec router ip addr add 10.0.1.2/30 dev veth-r1
sudo ip netns exec router ip addr add 10.0.2.1/30 dev veth-r2
sudo ip netns exec host2 ip addr add 10.0.2.2/30 dev veth-h2
daddy@DESKTOP-QFLJ22N:~$ sudo ip netns exec router ip addr add 10.0.1.2/30 dev veth-r1
daddy@DESKTOP-QFLJ22N:~$ sudo ip netns exec router ip addr add 10.0.2.1/30 dev veth-r2
daddy@DESKTOP-QFLJ22N:~$ sudo ip netns exec host2 ip addr add 10.0.2.2/30 dev veth-h2
sudo ip netns exec host2 ip addr add 10.0.2.2/30 dev veth-h2
daddy@DESKTOP-QFLJ22N:~$ sudo ip netns exec host1 ip link set veth-h1 up
daddy@DESKTOP-QFLJ22N:~$ sudo ip netns exec router ip link set veth-r1 up
daddy@DESKTOP-QFLJ22N:~$ sudo ip netns exec router ip link set veth-r2 up
daddy@DESKTOP-QFLJ22N:~$ sudo ip netns exec router ip link set veth-r3 up
Cannot find device "veth-r3"
daddy@DESKTOP-QFLJ22N:~$ sudo ip netns exec host2 ip link set veth-h2 up
daddy@DESKTOP-QFLJ22N:~$ sudo ip netns exec host1 ip link set lo up
daddy@DESKTOP-QFLJ22N:~$ sudo ip netns exec router ip link set lo up
daddy@DESKTOP-QFLJ22N:~$ sudo ip netns exec host2 ip link set lo up
daddy@DESKTOP-QFLJ22N:~$ sudo ip netns exec router sysctl -w net.ipv4.ip_forward=1
net.ipv4.ip_forward = 1
daddy@DESKTOP-QFLJ22N:~$ sudo ip netns exec host1 ip route add 10.0.2.0/30 via 10.0.1.2
daddy@DESKTOP-QFLJ22N:~$ sudo ip netns exec host2 ip route add 10.0.2.0/30 via 10.0.1.2
Error: Nexthop has invalid gateway.
daddy@DESKTOP-QFLJ22N:~$ sudo ip netns exec host2 ip route add 10.0.1.0/30 via 10.0.2.1
daddy@DESKTOP-QFLJ22N:~$ sudo ip netns exec host1 ping -c 3 10.0.2.2
PING 10.0.2.2 (10.0.2.2) 56(84) bytes of data.
64 bytes from 10.0.2.2: icmp_seq=1 ttl=63 time=8.74 ms
64 bytes from 10.0.2.2: icmp_seq=2 ttl=63 time=0.145 ms
64 bytes from 10.0.2.2: icmp_seq=3 ttl=63 time=0.092 ms

--- 10.0.2.2 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 1998ms
rtt min/avg/max/mdev = 0.092/2.993/8.742/4.065 ms
daddy@DESKTOP-QFLJ22N:~$ sudo ip netns exec router tc qdisc add dev veth-r2 root netem delay 20ms loss 1%
daddy@DESKTOP-QFLJ22N:~$ sudo ip netns exec router tc qdisc show dev veth-r2
```

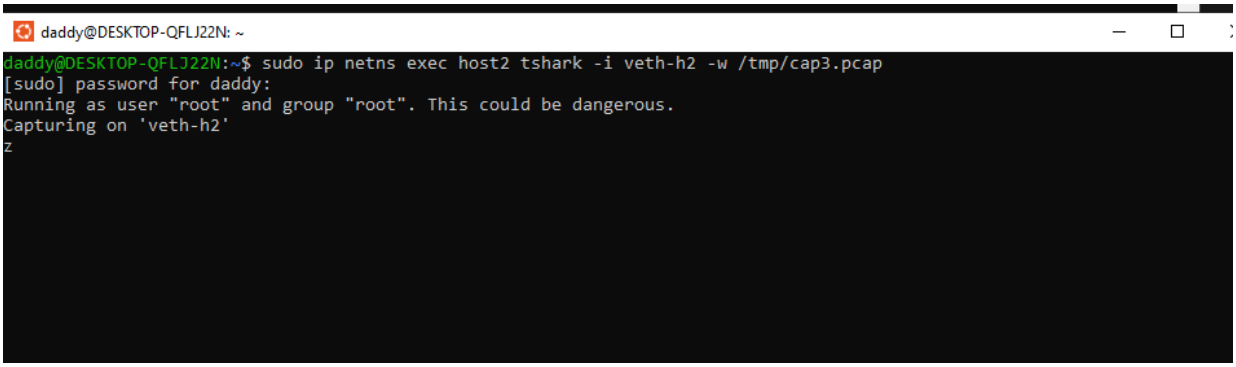
Terminal 1 - IP addresses assigned, interfaces up, routing enabled, ping test host1 to host2 successful



Terminal 1 - netem delay 20ms loss 1% applied on router and Capture 1 started on host1 (veth-h1)



Terminal 2 - Capture 2 started on router (veth-r1)



Terminal 3 - Capture 3 started on host2 (veth-h2)

Screenshot 1 – Namespaces and interfaces created

```
daddy@DESKTOP-QFLJ22N:~$ sudo ip netns list
host2
host1
daddy@DESKTOP-QFLJ22N:~$ sudo ip netns add router
sudo ip netns list
daddy@DESKTOP-QFLJ22N:~$ sudo ip netns list
router
host2
host1
```

Fig 1a: sudo ip netns list - namespaces host1, router, host2 created

```
daddy@DESKTOP-QFLJ22N:~$ sudo ip link set veth-r1 netns router
daddy@DESKTOP-QFLJ22N:~$ sudo ip link set veth-r2 netns router
daddy@DESKTOP-QFLJ22N:~$ sudo ip link set veth-h2 netns host2
daddy@DESKTOP-QFLJ22N:~$ sudo ip link show
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN mode DEFAULT group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc mq state UP mode DEFAULT group default qlen 1000
    link/ether 00:15:5d:b7:ed:08 brd ff:ff:ff:ff:ff:ff
daddy@DESKTOP-QFLJ22N:~$
```

Fig 1b: sudo ip link show - all veth interfaces moved into their namespaces (only lo and eth0 remain in the main namespace)

Screenshot 2 – Ping test host1 to host2

```
daddy@DESKTOP-QFLJ22N:~$ sudo ip netns exec host1 ping -c 3 10.0.2.2
PING 10.0.2.2 (10.0.2.2) 56(84) bytes of data.
64 bytes from 10.0.2.2: icmp_seq=1 ttl=63 time=8.74 ms
64 bytes from 10.0.2.2: icmp_seq=2 ttl=63 time=0.145 ms
64 bytes from 10.0.2.2: icmp_seq=3 ttl=63 time=0.092 ms

--- 10.0.2.2 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 1998ms
rtt min/avg/max/mdev = 0.092/2.993/8.742/4.065 ms
```

Fig 2: sudo ip netns exec host1 ping -c 3 10.0.2.2 - 3 packets transmitted, 3 received, 0% packet loss, ttl=63

Screenshot 3 – netem delay and loss applied on router

```
daddy@DESKTOP-QFLJ22N:~$ sudo ip netns exec router tc qdisc show dev veth-r2
qdisc netem 8001: root refcnt 3 limit 1000 delay 20ms loss 1%
daddy@DESKTOP-QFLJ22N:~$
```

Fig 3: sudo ip netns exec router tc qdisc show dev veth-r2 - qdisc netem delay 20ms loss 1%

Screenshot 4 – TTL check of cap1, cap2, cap3

```
      6 63
      8 64
daddy@DESKTOP-QFLJ22N:~$
```

Fig 4a: TTL in cap1 (host1) - 8 packets with TTL 64 (sent by host1), 6 packets with TTL 63 (replies from host2)

```
daddy@DESKTOP-QFLJ22N:~$ tshark -r /tmp/cap2.pcap -T fields -e ip.ttl -Y "ip" | sort | uniq -c
      6 63
      8 64
```

Fig 4b: TTL in cap2 (router) - 8 packets with TTL 64, 6 packets with TTL 63

```
daddy@DESKTOP-QFLJ22N:~$ tshark -r /tmp/cap3.pcap -T fields -e ip.ttl -Y "ip" | sort | uniq -c
      8 63
      6 64
daddy@DESKTOP-QFLJ22N:~$
```

Fig 4c: TTL in cap3 (host2) - 8 packets with TTL 63 (host1 packets after the router reduced the TTL by 1), 6 packets with TTL 64 (sent by host2)

Screenshot 5 – Message received on host2

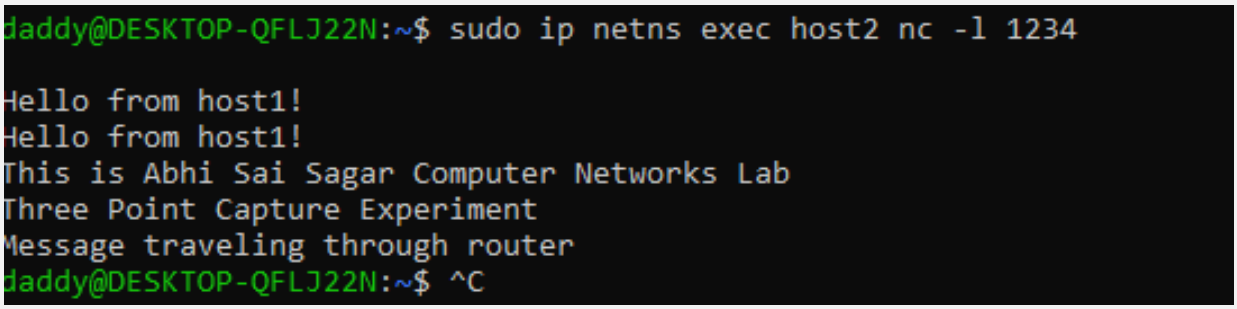


Fig 5a: Message received on host2 (Terminal 4, receiver: nc -l 1234) - the lines sent by host1 are displayed on host2

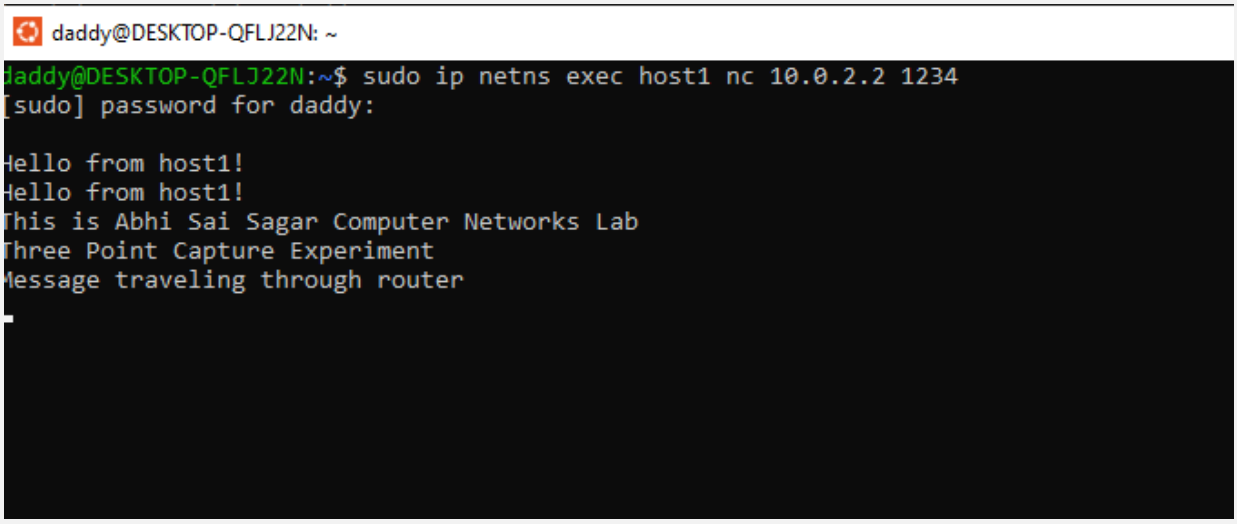


Fig 5b: Message sent from host1 (Terminal 5, sender: nc 10.0.2.2 1234) - "Hello from host1!" / "This is Abhi Sai Sagar Computer Networks Lab" / "Three Point Capture Experiment" / "Message traveling through router"

Conclusion

In this experiment, a two-subnet routed network (host1 - router - host2) was built entirely with Linux network namespaces and veth pairs, and one TCP conversation was captured simultaneously at three points of its path. Comparing the three captures shows what changes and what stays the same as a packet crosses a router: the payload and the port numbers are unchanged end-to-end (the exact message typed at host1 was read back from the router capture and displayed at host2), the TTL drops from 64 to 63 because the router decrements it by 1 while forwarding, the arrival time at host2 is about 20 ms later because of the netem delay on the router-to-host2 link, and retransmissions are concentrated at the host2 capture point because the 1% loss was applied on that final link. Three-point capture is therefore a powerful diagnostic method: by comparing the same flow at several points, the exact hop at which delay or loss is introduced can be located.